

CHE 317: INTRODUCTION TO CHEMICAL ENGINEERING ANALYSIS

Unique Number: 17000

Instructor:

Dr. Hal Alper

Teaching Assistant:

Celeste Marter

Teaching Modality Information:

This is an in-person course and class meetings will be held in-person throughout the semester.

Lectures:

T, Th 9:30 – 11:00 AM (CPE 2.216)

Please be punctual (lectures start at 9:30 AM PROMPTLY!). Lectures will be based on reading assignments that you should complete prior to coming to class. Lecture time will be used to both cover conceptual topics and work through in-class problem solving skills.

Recitation:

W 3 – 5 PM (CPE 2.216)

Please be punctual. Attendance is strongly encouraged as there will be weekly quizzes covering topics from homework assignments and lecture material. Your TA will be going over practice problems and answering questions at these sessions. Exams will also occur during these 2 hour recitation periods.

Course Goals and Overview:

Material and Energy Balances form the foundation for nearly all future chemical engineering courses and analysis. The process of balancing mass and energy is akin to “chemical accounting” and in this course we will learn how to formulate the balances that govern systems in which mass and energy are exchanged. Included in this analysis are both reactive and non-reactive systems. An overview of physical behavior will be included to provide an introduction to energy balances. Overall, this course will provide an introduction to the principles and calculation techniques used in the field of Chemical Engineering as well as provide an exposure of the various areas and facets of current Chemical Engineering research. This course is designed to:

- Provide you with the fundamentals of material and energy balances as applied to chemical engineering to enable the analysis of a chemical process
- Develop efficient methods of and life-long skills for problem solving through exercise problems and thought experiments
- Offer practice in defining problems, collecting data, and analyzing data
- Help you decide if you have chosen the right field

Course Prerequisites:

- One year of college-level mathematics (algebra, solutions of simultaneous equations, quadratic equations, basic differentiation and integration, trigonometry and basic geometry [areas and volumes of simple shapes]) – through *M408D*

- Chemistry (stoichiometry, concept of mole, molecular weight, concentration, molarity, molality, etc.; enthalpy of reaction, ideal gas law, pressure, absolute temperature) – *Ch302*
- Basic knowledge of statics and dynamics in physics (concepts of force, work, pressure, kinetic energy, potential energy) – *Phy303* or high school physics
- Spreadsheet calculations for solving systems of linear equations, function minimization, and plotting – *ChE210*
- Time management skills to balance course load and reading assignments

Course Textbook:

Elementary Principles of Chemical Processes, 4th Edition, Richard M Felder, Ronald W Rousseau, and Lisa G Bullard, ISBN-13: 978-1118431221

Course Website:

A course page will be maintained on Canvas—Please check it regularly! Announcements will be posted and sent out via Canvas.

Grading:

You may select one of two grading schemes [choice should be indicated on the last day of class]:

	<u>Scheme A</u>		<u>Scheme B</u>
Homework	10	Homework	10
Quizzes (lowest 2 dropped)	20	Quizzes (lowest 2 dropped)	20
Exams	70	Exams (Best 3 of 4)	52.5
Final Exam	0	Final Exam	17.5

Note that grades letter grades will be assigned based on natural breaks in the distribution. Plus/Minus grading will be used in this course.

Homework:

Homework will generally be assigned at the end of the lecture period on Tuesdays; they are due at the beginning of the specified lecture period (usually a week from assigned date). Late homework will not be accepted. Your homework must be neatly presented in the standard chemical engineering format. Homework must be handed in individually, but you are encouraged to work in groups to discuss and solve the problems. Homework problems are graded with an emphasis on effort and setting up the problem, and only selected problems within each assignment will be graded. A homework formatting guide will be provided with the first assignment, please follow it for all homework.

Reading Assignments:

At the start of each lecture, you will also be given a reading assignment. Read the sections prior to the next lecture. Work the example problems in your textbook as a guide to whether you have fully understood the material.

Quizzes:

There will be weekly quizzes held during the recitation period (may be either at the start or end of the period). These quizzes are based on materials covered in reading assignments, classroom discussions, and homework. Quizzes will not be held during the weeks of Exams so will result in

approximately 8-9 quizzes through the semester. You will be allowed to drop two of the lowest quiz grades this semester, which may be a missed quiz or your lowest grade. As a result, no make-up quizzes will be allowed. You have one week after the quiz is returned to you to request for re-grading. Re-grading requests should be submitted in writing at the end of the lecture period.

Exams:

Four exams are scheduled during **Wednesday** recitation period on: **September 16, October 7, October 28, November 18. There will be no make-up exams offered as grading scheme B allows for a dropped exam!** You have one week after the exam is returned to you to request for re-grading. Re-grading requests should be submitted in writing at the end of that lecture period. Re-grade requests should consist of a few sentences on a separate page that describe your rationale for why a re-grade is necessary. Optional review sessions (typically outside of class time) will be held before each of the exams. Additional problems for practice are always available from the instructor. The exam location will be announced both in class and on canvas in advance of the Exam date.

Final Exam:

The final exam (depends on grading scheme selected) is currently scheduled for **Monday, December 14, 8:00 AM-10:00 AM.**

Absence Policy:

No excuses for missed tests, exams, or quizzes will be accepted other than written certified medical excuses or written letters on university letterhead for UT-related school activities. By UT Austin policy, you must notify me of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Statement on Flexibility:

This course will reaffirm one of the core values here at UT Austin: responsibility. Our responsibility to ourselves and each other is to put our humanity in the forefront of our academic pursuits. If you experience any hardships such as illness, accident, family crisis please know that these policies may be amended and therefore you should communicate with me as soon as you feel comfortable doing so. If for any reason you do not feel comfortable discussing with me, please visit Student Emergency Services.

Statement on Learning Success:

Your success in this class is important to me. We will all need to be adaptable because we all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. Together we'll develop strategies to meet both your needs and the requirements of the course. I also encourage you to reach out to the student resources available through UT. Many are listed on this syllabus, but I am happy to connect you with a person or Center if you would like.

Knowledge, Abilities, and Skills Students Should Gain From this Course:

- Familiarity with chemical process equipment
- Exposure to systems analysis and concepts of unit operations
- Ability to convert between different sets of units, to use the appropriate gravitational conversion factor, and to distinguish between gauge and absolute pressures
- Ability to write and balance chemical reaction equations
- Ability to identify the excess and limiting reagents, and calculate the percent conversion and yield of reactions
- Ability to define steady state and distinguish between open and closed systems
- Ability to solve material balances for systems with or without a chemical reaction; ability to solve a number of interconnected units involving recycle, bypass, and purge
- Ability to define an ideal gas, and calculate based on the perfect gas equation
- Ability to apply Dalton's and Amagat's Laws for ideal gas mixtures
- Ability to define vapor pressure, saturation, equilibrium, triple point, dew point, partial saturation (humidity), etc.; ability to distinguish between wet and dry bulb temperatures
- Ability to apply the phase rule
- Ability to define heat, work, energy, enthalpy, etc.
- Ability to estimate heat capacities and calculate enthalpy changes, with and without phase changes
- Ability to use steam tables
- Ability to solve energy balances for open and closed systems, with or without chemical reactions
- Ability to solve simple combined material and energy balances (with or without chemical reactions)
- Ability to distinguish and calculate adiabatic and non-adiabatic problems
- Ability to apply spreadsheet calculations to mass and energy balance problems
- Exposure to library and reference data sources
- Ability to work in groups

ABET Student Outcomes for this course:

This course addresses the following ABET student outcomes:

Student Outcome 1: Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

Impact on Subsequent Courses in Curriculum:

Virtually every subsequent ChE course requires material and energy balance concepts. In addition, properties of materials (gas laws, vapor pressures, etc.) are important in *ChE319*, *ChE322*, *ChE354*, *ChE350*, *ChE 363*, *ChE372*, and *ChE473K*.

Academic Integrity:

Students who violate University rules on academic misconduct are subject to the student conduct process. All suspected misconduct will be referred to Student Conduct and Academic Integrity (SCAI) in the Office of the Dean of Students. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at:

<http://deanofstudents.utexas.edu/conduct>.

In particular, students are obliged to follow the Student Honor Code that states “As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity.”

Classroom Accessibility and Inclusivity:

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability & Access (D&A). Please refer to the D&A website for more information: <http://diversity.utexas.edu/disability/>. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

Important Safety Information:

If you have concerns about the safety or behavior of fellow students, TAs or Professors, call BCAL (the Behavior Concerns Advice Line): 512-232-5050. Your call can be anonymous. If something doesn't feel right – it probably isn't. Trust your instincts and share your concerns.

The following are recommendations regarding emergency evacuation from the Office of Emergency Management:

- Students should sign up for Campus Emergency Text Alerts.
- Occupants of buildings on The University of Texas at Austin campus must evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.
- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors. Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.

For more information, please visit the Office of Emergency Management (<https://emergencymanagement.utexas.edu/>).

Health is also safety!

Most importantly, you should stay home if you are sick or contagious, not only to stop the spread of disease, but also to promote your own personal wellness.

Title IX Reporting:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working

environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can:

1. Intervene to prevent harmful behavior from continuing or escalating.
2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation.
3. Investigate and discipline violations of the university's [relevant policies](https://titleix.utexas.edu/policies) (<https://titleix.utexas.edu/policies>).

Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. **I am a Responsible Employee and must report any Title IX related incidents** that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Sharing of Course Materials is Prohibited:

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students.

Classroom Recording:

HOP 2-9970 prohibits students from recording class instruction (audio or video) unless a student obtains the instructor's permission or Disability & Access has approved audio recording as an accommodation.

Course Artificial Intelligence Policy:

Generative AI use shall be permitted on a partial basis for academic work in this course during clearly marked, specific assignments aimed to evaluate AI for solving material and energy balances. Outside of these specific assignments, the use of Generative AI is NOT permitted and the use of generative AI tools for academic work may constitute academic misconduct under UT Austin's Institutional Rules and may be referred to Student Conduct and Academic Integrity in the Office of the Dean of Students for resolution.

Course content outline (as of August 25)

Note: Changes to the schedule and/or course modality may be made if circumstances require. It is your responsibility to note these changes when announced in class and via announcements on canvas. As much advanced notice will be provided as possible should anything change.

Part 1: Introduction to Engineering Analysis (Review on own: from 210)		
	Dimensions, units, and conversions	Chapter 2.1-2.5b, 2.6
	Process variables	Chapter 3.1-3.3
	Basis sets	Chapter 3.1-3.3
Part 2: Material Balances		
	Basic material balances (Review on own: from 210)	Chapter 4
	Problem solving definition and strategies (Review on own: from 210)	Chapter 4.1-4.2
	Material balances for single, unreactive units	Chapter 4.3
	Material balances accounting for reactions	Chapter 4.6
	Material balances for systems with reactions	Chapter 4.6-4.8
	Multi-unit processes with reactions	Chapter 4.4
	Material balances with recycle, bypass and purge	Chapter 4.5
Part 3: Behavior of Gases, Vapors, Liquids, and Solids		
	Temperature units and measurement	Chapter 3.5
	Pressure measurement and manometer applications	Chapter 3.4
	Ideal gases and mixtures with applications to material balances	Chapter 5.1-5.2
	Vapor pressure: Antoine, cox charts and steam tables	Chapter 5.1-5.2
	Saturation, condensation and vaporization	Chapter 6.1-6.5
	Partial saturation and humidity	Chapter 6.1-6.5
Part 4: Energy Balances		
	Types of energy and related concepts (potential, internal, kinetic and enthalpy changes, use of steam tables)	Chapter 7
	Basic non-reactive energy balances components	Chapter 7
	Energy balances without reactions	Chapter 8
	Combined mass and energy balances with reactions	Chapter 9
	Applications and analysis of combined mass and energy balances	Chapter 9
Part 5: Transient Balances		
	Balances on Transient Processes	Chapter 10